# A little book of very hard things

Tungsten Carbide

Name _____ Date _____



A gray tip rested on a cloth. “Could that little piece do a big job?” asked Little Wonderer. “It is tungsten carbide,” said Big. “A material is what something is made of. This one is heavy for its size and very hard. But being hard does not mean it can never break.”



Little found another clue in a pen. Many ballpoint pens have a tiny tungsten-carbide ball that rolls ink onto paper. People make carbide by joining tungsten, a metal, with carbon. They press the powder and heat it very hot. A little cobalt metal usually holds the grains together, like glue.



Big set soap and a wooden craft stick on a tray. Little guessed what would happen, then watched. Big gently drew the stick across the soap. A scratch appeared: a small line dug into the surface. Hard means resisting scratches and dents. The soap gave way to the harder stick.

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|---|---|
| **hard** | able to resist scratches and dents |
| **tough** | able to resist cracking or breaking from bumps and bends |
| **brittle** | able to chip or crack instead of bending much |
| **scratch** | a small line dug into a surface |
| **tip** | the small end of something |
| **material** | what something is made of |
| **tungsten** | a heavy metal used to make tungsten carbide |
| **carbon** | an element that joins with tungsten to make tungsten carbide |

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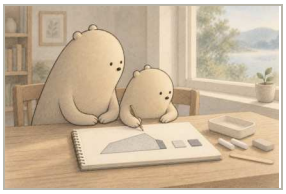
Name _____ Date _____



Does every material bend before it breaks? Big gently bent chalk over the tray. It snapped: chalk is brittle. A rubber eraser bent and came back: it is tough. Tough means resisting cracks and breaks from bumps or bends. Tungsten carbide is very hard, but brittle, so a strong knock can chip it.



On a snowy walk, Little noticed the tips of their hiking poles. The tip is the small end that meets the ground. Some pole tips and ice-grip studs use carbide to bite into ice. “The hard part is only a tiny piece!” Little said. “Just where it is useful,” said Big.



Back at the notebook, Big showed a saw-tooth picture. A small carbide tip joined a steel body. The hard tip resists wear; the tougher steel takes bumps without cracking as easily. “Two materials, two jobs,” Little said. “Do I need it hard, tough, or both?”

1. What is the difference between hard and tough?
2. What happened to the chalk and the eraser when Big bent them?
3. Why can a carbide tip and a steel body be a useful pair?

## Your turn

Write one command for the soap test.

Example: Ask Big to hold the soap and gently drag the craft stick across it.